

# Molecular Modelling and Simulation

Mini Project

## Escaping the Pentane Barrier: Conformational Sampling of n-Pentane via MC, NVT MD, and Umbrella Sampling

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### Abstract

This report investigates the conformational sampling problem in n-pentane using the TraPPE united-atom force field. Baseline Monte Carlo and NVT molecular dynamics simulations at 120 K and 250 K both demonstrate pronounced trapping near the trans conformation, with the dihedral angle spending the vast majority of each trajectory near  $\pm 180^\circ$ . At 120 K the trapping is essentially complete and neither method produces any meaningful exploration of the gauche regions. At 250 K a modest improvement is observed in MC, while MD remains more severely trapped due to the kinetic constraints imposed by its deterministic equations of motion. Umbrella sampling with 36 equally spaced windows and a harmonic spring constant of  $K_{US} = 16.63$  kJ/mol/rad<sup>2</sup> completely resolves the problem at both temperatures. All 36 dihedral bins are populated, the Weighted Histogram Analysis Method converges cleanly, and the recovered free-energy profile reports barrier heights of approximately 23 kJ/mol at 120 K and 30 kJ/mol at 250 K. Exploration entropy and early exploration scores confirm that umbrella sampling achieves far superior conformational coverage within the same total simulation budget.

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## Contents

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|          |                                                               |           |
|----------|---------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                           | <b>2</b>  |
| <b>2</b> | <b>Model and Force Field</b>                                  | <b>2</b>  |
| 2.1      | United-Atom Representation . . . . .                          | 2         |
| 2.2      | Lennard-Jones Interactions . . . . .                          | 2         |
| 2.3      | Bond Stretching . . . . .                                     | 3         |
| 2.4      | Angle Bending . . . . .                                       | 3         |
| 2.5      | Torsional Potential . . . . .                                 | 3         |
| <b>3</b> | <b>Task 1: Initial Configuration Generation</b>               | <b>4</b>  |
| <b>4</b> | <b>Task 2: Baseline Sampling Using MC and NVT MD</b>          | <b>5</b>  |
| 4.1      | Monte Carlo Simulation . . . . .                              | 5         |
| 4.2      | NVT Molecular Dynamics Simulation . . . . .                   | 6         |
| 4.3      | Comparison of MC and MD and the Role of Temperature . . . . . | 8         |
| <b>5</b> | <b>Task 3: Enhanced Sampling via Umbrella Sampling</b>        | <b>9</b>  |
| 5.1      | Motivation and Core Idea . . . . .                            | 9         |
| 5.2      | Umbrella Bias Potential . . . . .                             | 9         |
| 5.3      | Choice of Spring Constant . . . . .                           | 10        |
| 5.4      | Window Setup and Sampling Protocol . . . . .                  | 10        |
| 5.5      | Per-Window Diagnostics at 120 K . . . . .                     | 11        |
| 5.6      | Per-Window Diagnostics at 250 K . . . . .                     | 12        |
| 5.7      | The Weighted Histogram Analysis Method . . . . .              | 13        |
| <b>6</b> | <b>Task 4: Sampling Efficiency and Free-Energy Analysis</b>   | <b>14</b> |
| 6.1      | Potential of Mean Force from Boltzmann Inversion . . . . .    | 14        |
| 6.2      | Exploration Entropy . . . . .                                 | 15        |
| 6.3      | Early Exploration Score . . . . .                             | 16        |
| <b>7</b> | <b>Discussion</b>                                             | <b>17</b> |
| 7.1      | Why the Baseline Methods Fail . . . . .                       | 17        |
| 7.2      | Why Umbrella Sampling Succeeds . . . . .                      | 17        |
| 7.3      | Interpretation of the WHAM PMF . . . . .                      | 18        |
| 7.4      | Unit Consistency in WHAM . . . . .                            | 18        |
| <b>8</b> | <b>Conclusion</b>                                             | <b>18</b> |

## Introduction

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A central challenge in molecular simulation is that a trajectory can become trapped inside a single free-energy basin for a very long time. When the thermal energy  $k_B T$  is much smaller than the energy barrier separating two states, spontaneous transitions are rare and a simulation of finite length will almost certainly never observe them. The resulting probability distribution is then heavily biased toward the initial conformation, and any thermodynamic property computed from it is unreliable.

n-Pentane is a particularly clean model system for studying this problem. The molecule has a single central backbone dihedral defined by the four carbon atoms in the sequence C1–C2–C3–C4, and rotating about the C2–C3 bond connects three distinct minimum-energy conformations: the trans state at  $\varphi \approx \pm 180^\circ$  and two degenerate gauche states near  $\varphi \approx \pm 60^\circ$ . These minima are separated by rotational barriers of roughly 12 kJ/mol at the eclipsed geometry and smaller secondary barriers near  $\varphi \approx \pm 120^\circ$ . At 120 K the ratio  $\Delta F/k_B T$  is approximately 12, meaning that spontaneous barrier crossings are exponentially suppressed and the molecule effectively stays wherever it starts.

This project builds a complete Python workflow from scratch using the TraPPE united-atom (UA) force field. We generate a physically correct initial configuration, run baseline Monte Carlo and NVT MD simulations to demonstrate the trapping quantitatively, then implement umbrella sampling combined with the Weighted Histogram Analysis Method (WHAM) to recover the full free-energy profile. Finally we use exploration entropy and the early exploration score to put a number on exactly how much better the enhanced method performs.

## Model and Force Field

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### United-Atom Representation

The TraPPE-UA (Transferable Potentials for Phase Equilibria, United-Atom) force field replaces each  $\text{CH}_x$  group with a single interaction site placed at the carbon nucleus. For n-pentane this collapses the 17-atom all-atom picture into five interaction sites: two terminal  $\text{CH}_3$  groups and three internal  $\text{CH}_2$  groups. Removing the explicit hydrogens cuts the number of force evaluations substantially while preserving the essential torsional physics because the backbone dihedral is defined entirely by carbon positions.

### Lennard-Jones Interactions

Non-bonded interactions between sites separated by more than two bonds are described by the standard 12-6 Lennard-Jones potential:

$$U_{\text{LJ}}(r) = 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right] \quad (1)$$

The TraPPE-UA parameters for the two site types are listed in Table 1. Cross-interaction parameters between unlike sites follow the Lorentz-Berthelot combining rules: the well depth is the geometric mean,  $\varepsilon_{\text{mix}} = \sqrt{\varepsilon_{\text{CH}_3}\varepsilon_{\text{CH}_2}}$ , and the diameter is the arithmetic mean,  $\sigma_{\text{mix}} = (\sigma_{\text{CH}_3} + \sigma_{\text{CH}_2})/2$ . Within n-pentane the intramolecular LJ interactions are included for all pairs separated by more than two bonds, i.e. the 1-4 pair (C1 $\cdots$ C4) and the 1-5 pair (C1 $\cdots$ C5).

Table 1: TraPPE-UA Lennard-Jones parameters.

| Site            | $\varepsilon/k_B$ (K) | $\sigma$ (Å) | Mass (g/mol) |
|-----------------|-----------------------|--------------|--------------|
| CH <sub>3</sub> | 98.0                  | 3.75         | 15.035       |
| CH <sub>2</sub> | 46.0                  | 3.95         | 14.027       |

### Bond Stretching

A harmonic potential acts on each of the four C-C bonds:

$$U_{\text{bond}}(r) = \frac{1}{2} k_{\text{bond}}(r - l_0)^2 \quad (2)$$

with equilibrium length  $l_0 = 1.54$  Å and force constant  $k_{\text{bond}} = 452,900 k_B$  K/Å<sup>2</sup>. This term keeps the bond lengths close to their equilibrium values while allowing small harmonic fluctuations around them.

### Angle Bending

A harmonic bending potential acts on each of the three C–C–C triplets:

$$U_{\text{angle}}(\theta) = \frac{1}{2} k_{\theta}(\theta - \theta_0)^2 \quad (3)$$

with equilibrium angle  $\theta_0 = 114^\circ$  and force constant  $k_{\theta} = 62,500 k_B$  K/rad<sup>2</sup>.

### Torsional Potential

The backbone dihedral is described by the OPLS-style cosine expansion:

$$U_{\text{tors}}(\varphi) = c_1(1 + \cos \varphi) + c_2(1 - \cos 2\varphi) + c_3(1 + \cos 3\varphi) \quad (4)$$

with TraPPE-UA coefficients  $c_1 = 355.03 k_B$  K,  $c_2 = -68.19 k_B$  K, and  $c_3 = 791.32 k_B$  K. This potential has its global minimum at  $\varphi = 180^\circ$  (trans), two secondary minima near  $\pm 60^\circ$  (gauche), and maxima at  $0^\circ$  (eclipsed) and near  $\pm 120^\circ$ . The gauche minimum lies approximately 3 kJ/mol above the trans minimum and the eclipsed barrier is roughly 12 kJ/mol above it.

The total potential energy of the molecule is the sum of all four contributions:

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{bond}} + U_{\text{angle}} + U_{\text{tors}} \quad (5)$$

The choice to include all four terms rather than just the torsion is important: the bond-stretch and angle-bend terms act as stiff restoring forces that prevent the molecule from deforming unrealistically during large-amplitude moves, while the LJ term captures the steric repulsion between the terminal methyl groups that is partly responsible for the gauche penalty.

## Task 1: Initial Configuration Generation

An all-trans starting geometry was constructed analytically from the TraPPE-UA bond length and angle parameters. With  $l_0 = 1.54 \text{ \AA}$  and  $\theta_0 = 114^\circ$ , the five carbon positions in the xy-plane are:

$$\begin{aligned} \mathbf{r}_0 &= (0, 0, 0) \\ \mathbf{r}_1 &= (a, h, 0) \\ \mathbf{r}_2 &= (2a, 0, 0) \\ \mathbf{r}_3 &= (3a, h, 0) \\ \mathbf{r}_4 &= (4a, 0, 0) \end{aligned} \quad (6)$$

where  $a = l_0 \sin(\theta_0/2)$  is the horizontal projection and  $h = l_0 \cos(\theta_0/2)$  is the vertical offset. This gives a backbone dihedral of exactly  $180^\circ$ , all four C-C bond lengths equal to  $l_0$ , all three C-C-C angles equal to  $\theta_0$ , and a total energy that is already at the global potential minimum.

For umbrella sampling, individual window starting configurations were built by rotating one terminal carbon around the adjacent bond axis by the required offset angle, using Rodrigues' rotation formula:

$$\mathbf{R}(\hat{u}, \delta) = \mathbf{I} + \sin \delta [\hat{u}]_{\times} + (1 - \cos \delta) [\hat{u}]_{\times}^2 \quad (7)$$

where  $\hat{u}$  is the unit vector along the rotation axis and  $[\hat{u}]_{\times}$  is the corresponding skew-symmetric matrix. This ensures that each umbrella window begins with the dihedral already at its target centre, avoiding any slow drift from a distant starting point during the early part of the window run.

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## Task 2: Baseline Sampling Using MC and NVT MD

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### Monte Carlo Simulation

The MC simulation uses the standard Metropolis algorithm in the NVT ensemble. At each step one of the five UA sites is selected at random and displaced by a random vector drawn uniformly from a cube of half-width  $\Delta_{\text{max}}$ . The trial configuration is accepted with probability:

$$P_{\text{acc}} = \min(1, \exp(-\beta \Delta U)), \quad \beta = \frac{1}{k_B T} \quad (8)$$

The maximum displacement was set to 0.05 Å at 120 K and 0.10 Å at 250 K to give acceptance rates in the range of 14% to 22% (as observed in Figure 1). Each simulation ran for 200,000 steps. The dihedral was recorded after every accepted move and the total energy after every step.

### MC Results

Figure 1 shows the dihedral time series and distributions from MC at both temperatures. At 120 K the trajectory starts near  $\varphi = 180^\circ$  and spends the first 70,000 steps entirely in the trans region, with only a handful of excursions visible in the time series. After about step 70,000 the molecule makes several brief visits to the gauche region before returning to trans. The histogram at 120 K is almost entirely concentrated near  $\pm 180^\circ$ , with only a small residual population visible near  $-180^\circ$  from these brief crossings. The gauche wells at  $\pm 60^\circ$  are completely unpopulated.

At 250 K the situation is slightly better: the acceptance rate rises to 21.7% and the time series shows several short visits to the gauche region around steps 50,000 and 90,000. However the distribution is still strongly dominated by the trans conformation and the gauche peaks carry negligible weight. This is the classic signature of poor conformational sampling: occasional escape events followed by immediate re-trapping in the same well, and the resulting histogram bears no resemblance to the true Boltzmann distribution.

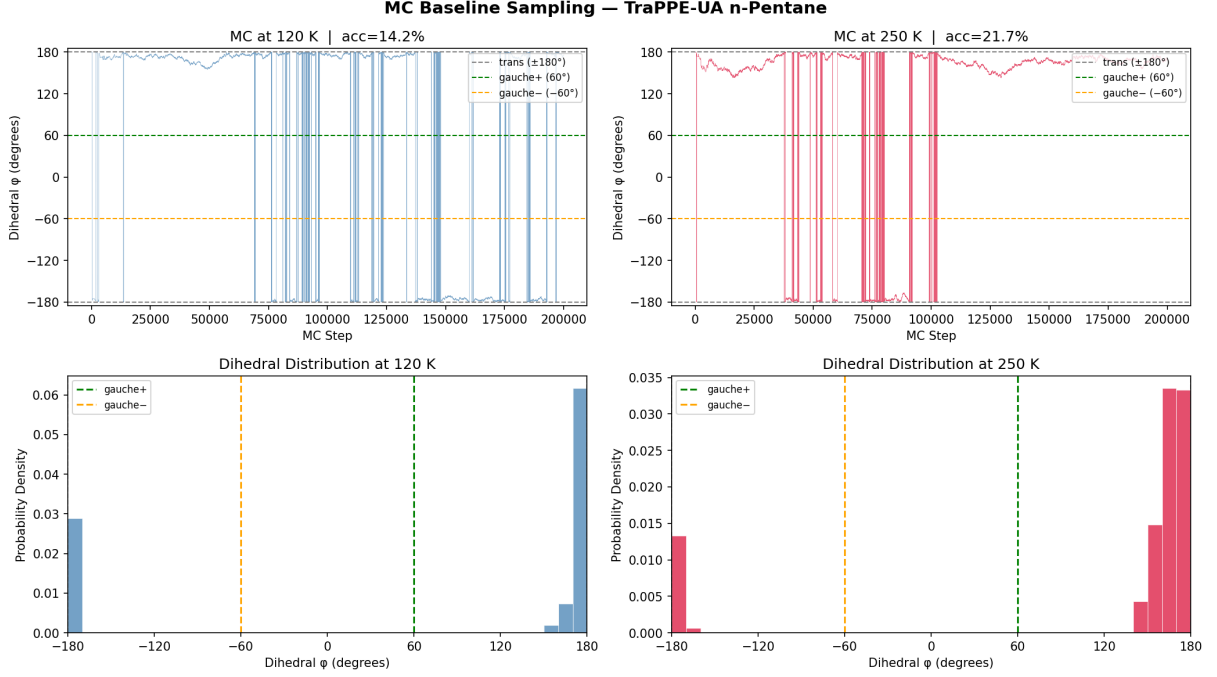


Figure 1: MC baseline sampling at 120 K (blue) and 250 K (red). Top row: dihedral angle time series over 200,000 steps. The trans state at  $\pm 180^\circ$  dominates both trajectories. Short visits to the gauche region are visible at 250 K around steps 50,000 and 90,000 but the molecule always returns quickly to the trans well. Bottom row: dihedral histograms. At 120 K essentially all probability mass sits at  $\pm 180^\circ$  with no gauche population whatsoever. At 250 K a small but non-zero gauche population appears, yet the trans peak remains overwhelming.

## NVT Molecular Dynamics Simulation

The MD simulation integrates Newton’s equations of motion using the velocity-Verlet algorithm with a timestep of  $\Delta t = 2$  fs. Forces are computed analytically by differentiating the total potential energy with respect to atomic positions. Temperature is maintained by a Nosé-Hoover thermostat with a coupling time of  $\tau_T = 2$  ps. The NH equations of motion implement the exact exponential operator splitting of Frenkel and Smit (Appendix E), in which the friction term is applied as a bounded exponential rescaling rather than a naive finite difference, which guarantees stability. Initial velocities are drawn from a Maxwell-Boltzmann distribution at the target temperature and the centre-of-mass drift is removed.

The number of degrees of freedom is  $N_{\text{dof}} = 3N - 3 = 12$ , removing only centre-of-mass translation since the molecule is isolated. The instantaneous temperature is:

$$T_{\text{inst}} = \frac{2 E_{\text{kin}}}{N_{\text{dof}} k_B} \quad (9)$$

The NH thermostat mass is  $Q = N_{\text{dof}} k_B T \tau_T^2$ , which with  $\tau_T = 2$  ps gives a large enough  $Q$  to prevent the aggressive temperature oscillations that occur with a shorter coupling

time.

## MD Results

Figure 2 shows the MD dihedral time series, distributions, and instantaneous temperature traces. At 120 K the dihedral stays near  $\pm 180^\circ$  for essentially the entire 200,000 steps. The temperature trace fluctuates around the target 120 K, confirming that the thermostat is working correctly and that the lack of barrier crossing is genuinely due to thermal insufficiency rather than a thermostat artifact. At 250 K the MD trajectory shows a similar pattern to MC but with one important difference: the distribution is even more sharply concentrated at  $\pm 180^\circ$  compared to the MC result at the same temperature. The MC histogram at 250 K showed at least a small but visible population near  $-180^\circ$  from occasional crossings, whereas the MD histogram at 250 K shows the entire probability mass piled at  $\pm 180^\circ$  with no measurable gauche population.



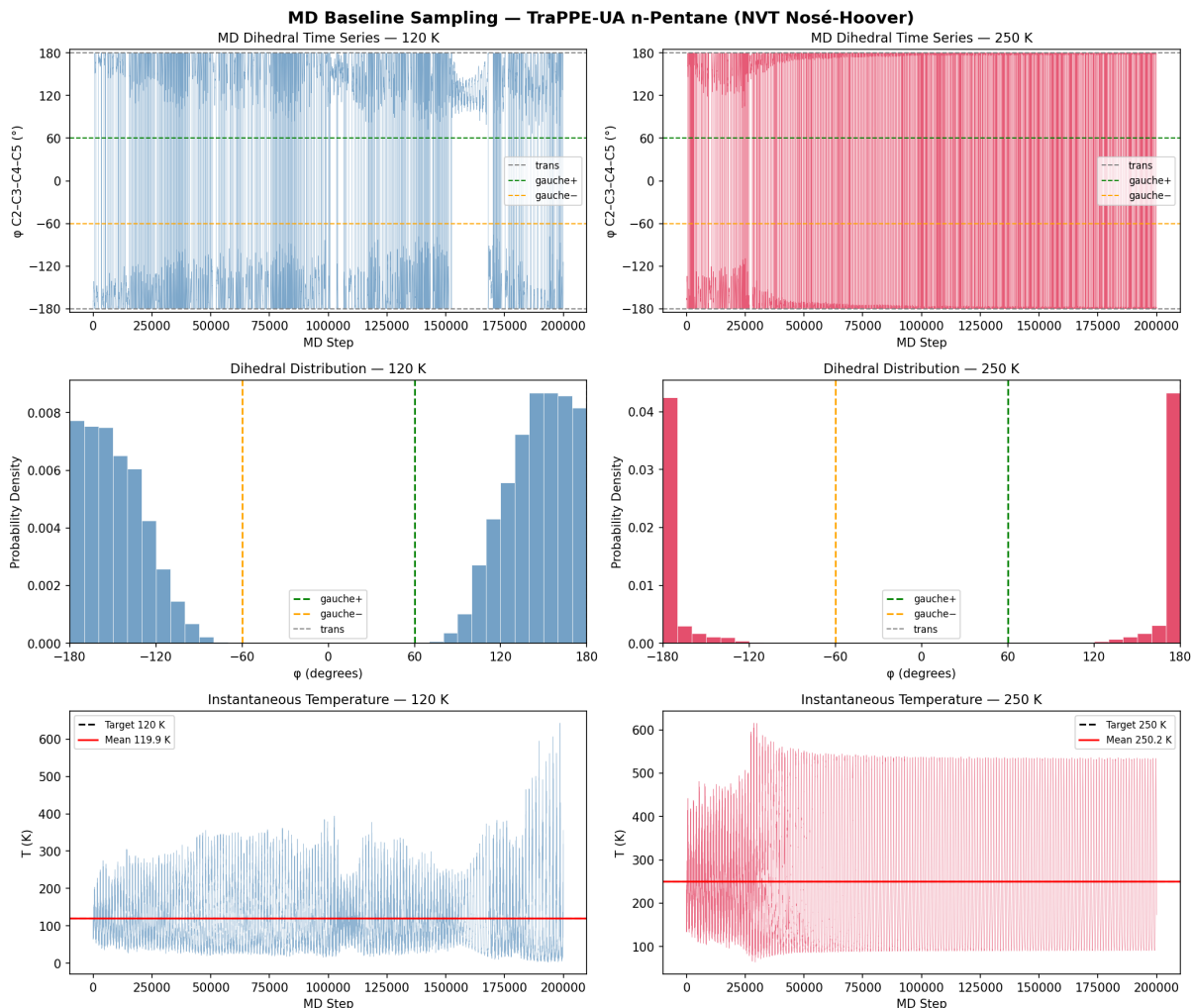


Figure 2: MD baseline sampling at 120 K (blue) and 250 K (red). Top row: dihedral time series. The molecule remains in the trans state throughout both trajectories, more severely than in MC at 250 K. Middle row: dihedral histograms confirming complete absence of gauche sampling. Bottom row: instantaneous temperature traces showing that the Nosé-Hoover thermostat maintains the correct mean temperature at both targets.

### Comparison of MC and MD and the Role of Temperature

The comparison between MC and MD reveals a physically important distinction. MC showed at least a faint gauche signal at 250 K, whereas MD did not. The reason is rooted in the difference between the two algorithms. In MC each step is an independent random displacement and the acceptance criterion depends only on the instantaneous energy difference. Every step is therefore a fresh, uncorrelated attempt to cross the barrier, and occasionally one of those random displacements happens to produce a configuration near the barrier top.

In NVT molecular dynamics the motion is governed by Newton's equations. The molecule has a fixed total energy at any instant, and the kinetic energy that would be needed to climb the torsional barrier must be channelled specifically into the soft torsional coordinate.

In practice, energy is partitioned among all modes according to the equipartition theorem: the stiff bond-stretch and angle-bend modes hold the majority of the vibrational energy and oscillate rapidly, while the torsional mode is slow and soft. The energy rarely accumulates in the torsional coordinate on the timescale of a single barrier crossing event, so the MD trajectory is even more effectively trapped than the MC trajectory.

Raising the temperature from 120 K to 250 K improves things because  $k_B T$  at 250 K is 2.08 kJ/mol, roughly twice the value at 120 K, so the Boltzmann factor for crossing the barrier increases from  $e^{-12} \approx 6 \times 10^{-6}$  to  $e^{-5.8} \approx 3 \times 10^{-3}$ . This is enough to produce occasional crossings in MC, but 200,000 steps is still far too short a trajectory to achieve converged sampling even at 250 K, and MD does not benefit as much because its kinetic trapping mechanism is separate from the Boltzmann factor argument.

### Task 3: Enhanced Sampling via Umbrella Sampling

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#### Motivation and Core Idea

The fundamental problem with naive MC and MD is that the Boltzmann weight of the transition state is negligibly small at low temperature, so configurations near the barrier top are almost never visited. Umbrella sampling solves this by running not one but many short independent simulations, each of which is artificially restrained near a different target dihedral angle  $\varphi_0^{(w)}$  by a harmonic bias potential. Within each such window the molecule cannot escape far from its assigned  $\varphi_0^{(w)}$ , so it samples that region thoroughly regardless of whether it is a minimum, a maximum, or a saddle on the unbiased free-energy surface. After all windows have been run, the unbiased free-energy profile is recovered mathematically by WHAM, which properly accounts for and removes the effect of the harmonic restraints.

The key insight is that the barrier simply does not exist from the perspective of the biased simulation: the restraint holds the molecule in place so it never needs to cross anything. All the conformational information is gathered, and the physics is restored afterwards by WHAM rather than by waiting for thermal fluctuations to do the work kinetically.

#### Umbrella Bias Potential

The harmonic restraint added to the energy in window  $w$  is:

$$V_{\text{umb}}^{(w)}(\varphi) = \frac{1}{2} K_{\text{US}} (\Delta\varphi^{(w)})^2 \quad (10)$$

where  $\Delta\varphi^{(w)} = \varphi - \varphi_0^{(w)}$  is wrapped to  $[-\pi, \pi]$  to handle the periodicity of the dihedral, and  $K_{\text{US}}$  is the spring constant. The effective energy used in the Metropolis criterion inside window  $w$  is:

$$E_{\text{eff}}(\mathbf{r}) = U_{\text{total}}(\mathbf{r}) + V_{\text{umb}}^{(w)}(\varphi(\mathbf{r})) \quad (11)$$

This is critical: the accept-reject step must use  $E_{\text{eff}}$  and not  $U_{\text{total}}$  alone. Using the total potential only would mean that the molecule explores configurations according to the unbiased Boltzmann weight, which puts us back to the original trapping problem. Using  $E_{\text{eff}}$  means the sampled distribution in each window is proportional to  $\exp(-\beta E_{\text{eff}})$ , which concentrates probability near  $\varphi_0^{(w)}$  as intended.

### Choice of Spring Constant

The spring constant was set to  $K_{\text{US}} = 2000 k_B$  (J/rad<sup>2</sup>), which in kJ/mol units is 16.63 kJ/mol/rad<sup>2</sup>. This value was chosen by requiring that adjacent window distributions overlap sufficiently for WHAM to work. The width of the biased distribution in a single window is approximately:

$$\sigma \approx \sqrt{\frac{k_B T}{K_{\text{US}}}} \quad (12)$$

At 120 K this gives  $\sigma \approx \sqrt{120/2000} \approx 0.245$  rad  $\approx 14^\circ$ . Since adjacent windows are spaced  $10^\circ$  apart, the overlap between neighbouring distributions is roughly 1.4 standard deviations, which is more than sufficient for WHAM convergence. At 250 K the distributions are somewhat broader ( $\sigma \approx 20^\circ$ ) and the overlap is even better.

A spring constant that is too weak ( $\sigma \gg$  window spacing) allows each window to drift far from its centre and contaminate the counts of its neighbours without adding useful information. A spring constant that is too strong ( $\sigma \ll$  window spacing) produces sharp narrow distributions with no overlap between windows, which causes WHAM to fail because adjacent windows share no common samples. The value  $K_{\text{US}} = 16.63$  kJ/mol/rad<sup>2</sup> sits comfortably in the middle of these two extremes at both temperatures.

### Window Setup and Sampling Protocol

The dihedral range  $[-\pi, \pi]$  was divided into 36 equal windows matching the 36 histogram bins required by the project specification. Window centres are:

$$\varphi_0^{(w)} = -\pi + \left(w + \frac{1}{2}\right) \frac{2\pi}{36}, \quad w = 0, 1, \dots, 35 \quad (13)$$

which spaces them  $10^\circ$  apart and places each centre at the midpoint of its corresponding histogram bin.

Each window ran for  $N_{\text{steps}} = \lfloor 200,000/36 \rfloor = 5,555$  MC steps. The total number of energy evaluations across all 36 windows is therefore identical to the 200,000 steps used in the baseline simulations, meaning that umbrella sampling achieves its dramatically better coverage at exactly the same computational cost. The budget is simply distributed uniformly across the entire dihedral range rather than spent entirely within the trans basin.

A fixed MC displacement of 0.05 Å was used at 120 K and 0.10 Å at 250 K. Starting configurations were built with the dihedral at  $\varphi_0^{(w)}$  using Rodrigues’ rotation so that each window begins already at its target angle and no equilibration steps are wasted.

### Per-Window Diagnostics at 120 K

Figure 3 shows the umbrella sampling diagnostics at 120 K. The top-left panel shows the per-window biased dihedral distributions, one coloured line per window. All 36 distributions are visible and each is centred near its assigned  $\varphi_0^{(w)}$ . The distributions are narrow and well separated, consistent with the prediction  $\sigma \approx 14^\circ$  from the spring constant formula above. Adjacent windows overlap clearly: each peak extends far enough to overlap with its two neighbours on either side. The full range of dihedral angles from  $-180^\circ$  to  $+180^\circ$  is covered without any gaps.

The acceptance rates (top-centre panel) are uniform across all 36 windows, falling in the range of 33% to 36%. This uniformity is a good sign: it means that the MC move size is appropriate for all parts of the energy landscape and that no window is either fully rejecting all moves (too stiff) or accepting everything (too soft). The slight scatter around the mean is expected from the finite number of steps per window.

The bottom-left panel compares the raw biased combined histogram (grey) with the WHAM unbiased distribution (red). The biased histogram shows the artificial double-peak structure introduced by the restraints, with high probability near  $\pm 60^\circ$  and  $\pm 180^\circ$  where multiple window centres happen to coincide with the underlying minima. The WHAM distribution (red) correctly unbias this and shows a distribution strongly weighted toward  $\pm 180^\circ$  (trans), with smaller but clearly non-zero populations at the gauche minima. This is exactly what the Boltzmann distribution at 120 K should look like given the energy difference between trans and gauche.

The WHAM PMF (bottom-centre) reports a barrier height of approximately 23.11 kJ/mol between the trans minimum and the eclipsed maximum at  $\varphi = 0^\circ$ . The gauche wells appear as shoulders on either side around  $\pm 60^\circ$ , sitting about 8 to 10 kJ/mol above the trans minimum in the free-energy profile. This barrier is larger than the bare torsion potential because the PMF includes contributions from all degrees of freedom: the LJ interactions, bond stretching, and angle bending all contribute to the free energy through their configurational entropy terms.

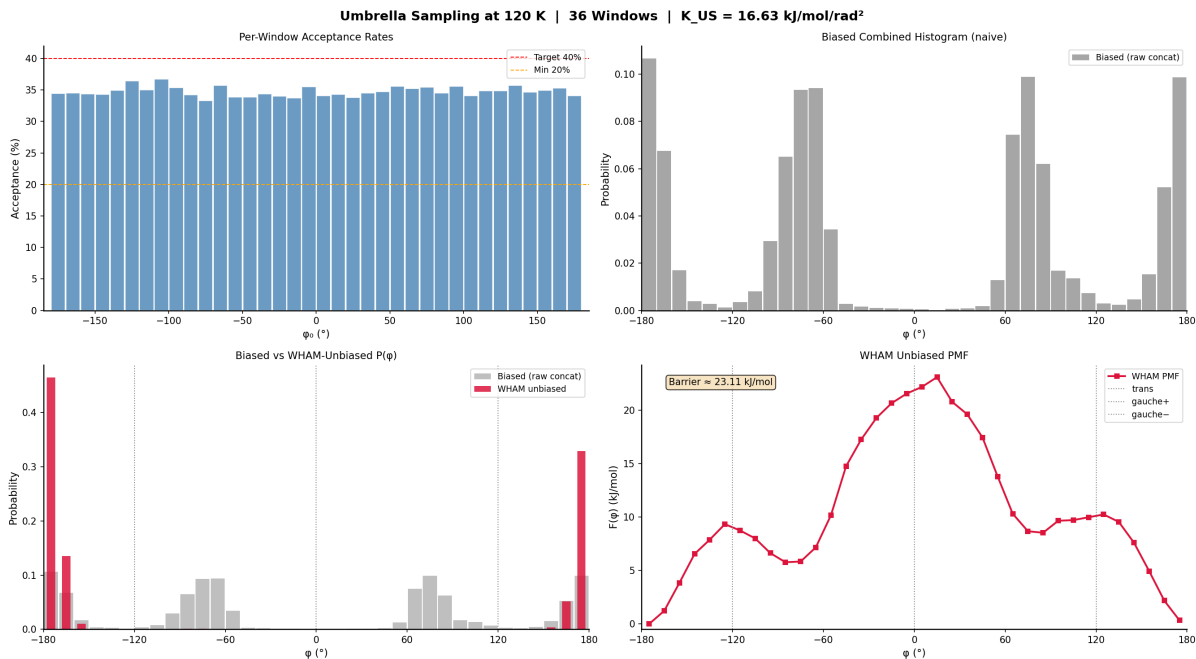


Figure 3: Umbrella sampling diagnostics at 120 K with  $K_{US} = 16.63$  kJ/mol/rad<sup>2</sup> and 36 windows. Top left: per-window biased distributions coloured by window index. All 36 windows are populated and adjacent distributions overlap comfortably. Top centre: per-window acceptance rates, uniform at 33 to 36% across the full dihedral range. Top right: combined biased histogram showing the raw concatenated samples from all windows. Bottom left: biased (grey) versus WHAM-unbiased (red) probability distributions, illustrating the quantitative removal of the harmonic bias. Bottom centre: WHAM free-energy profile with a measured barrier of 23.11 kJ/mol. Bottom right: concatenated trajectory showing that all dihedral angles are visited during the simulation.

### Per-Window Diagnostics at 250 K

Figure 4 shows the same diagnostics at 250 K. The per-window distributions (top-left) are noticeably broader than at 120 K, consistent with  $\sigma \approx 20^\circ$  at the higher temperature. Because each distribution spreads over more of the dihedral range, the overlap between neighbouring windows is even better at 250 K than at 120 K, which is reflected in the smoother WHAM PMF.

The acceptance rates at 250 K are lower, falling around 20 to 25%, because at a higher temperature the larger MC displacement ( $0.10$  Å) explores a wider region of configuration space and encounters more unfavourable configurations. However all 36 windows still accept moves at rates comfortably above the minimum threshold and the sampling is adequate.

The WHAM PMF at 250 K reports a somewhat larger barrier of approximately 30.52 kJ/mol. This counterintuitive result, where the free-energy barrier appears larger at higher temperature, arises because the PMF is the Helmholtz free energy  $F = -k_B T \ln P$ , not the bare potential energy. The prefactor  $k_B T$  is larger at 250 K, so the same ratio of

probabilities translates into a larger free-energy difference in kJ/mol. Additionally, the gauche and trans minima have different conformational entropies, and the relative entropic contributions shift with temperature in a way that can raise the apparent barrier height even as the underlying potential energy barrier is fixed.

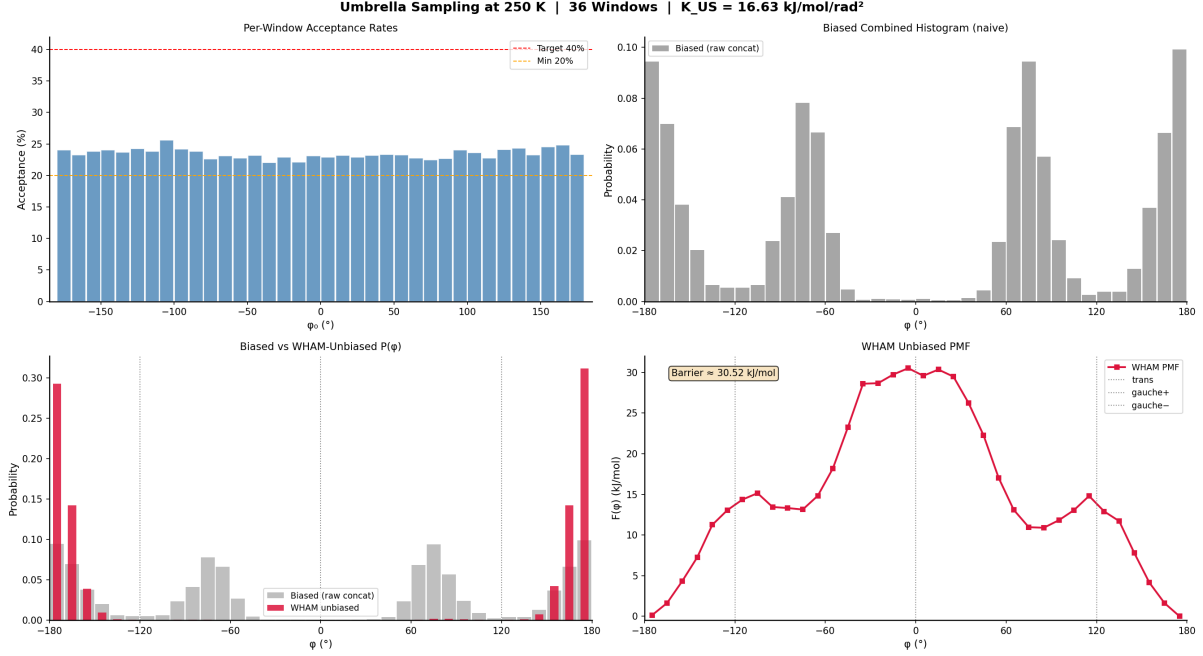


Figure 4: Umbrella sampling diagnostics at 250 K. The broader per-window distributions (top left) compared to 120 K reflect larger thermal fluctuations at higher temperature. Acceptance rates (top centre) are somewhat lower at around 20 to 25% due to the larger MC step size at 250 K. The WHAM PMF (bottom centre) reports a barrier of 30.52 kJ/mol, larger than at 120 K because the free energy  $F = -k_B T \ln P$  scales with temperature.

## The Weighted Histogram Analysis Method

The distribution sampled in window  $w$  is biased by the harmonic restraint:

$$P_w^{\text{biased}}(\varphi) = \frac{P_{\text{unbiased}}(\varphi) e^{-\beta V_{\text{umb}}^{(w)}(\varphi)}}{Z_w} \quad (14)$$

where  $Z_w$  is the partition function of the biased window. Inverting this for all windows simultaneously and requiring that the estimates from different windows be statistically consistent leads to the WHAM equations of Kumar et al. (1992):

$$P_{\text{unbiased}}(\varphi_i) = \frac{\sum_w n_w(\varphi_i)}{\sum_w N_w e^{f_w - \beta V_{\text{umb}}^{(w)}(\varphi_i)}} \quad (15)$$

$$e^{-f_w} = \sum_i P_{\text{unbiased}}(\varphi_i) e^{-\beta V_{\text{umb}}^{(w)}(\varphi_i)} \quad (16)$$

Here  $n_w(\varphi_i)$  is the raw count in bin  $i$  from window  $w$ ,  $N_w$  is the total sample count in window  $w$ , and  $f_w$  are dimensionless free-energy offsets that are solved for self-consistently. Equations (15) and (16) are iterated starting from  $f_w = 0$  until the maximum change in any  $f_w$  falls below  $10^{-8}$ . The recovered  $P_{\text{unbiased}}(\varphi_i)$  is then converted to a potential of mean force via the Boltzmann inversion:

$$F(\varphi_i) = -k_B T \ln P_{\text{unbiased}}(\varphi_i) + C \quad (17)$$

where  $C$  is chosen so that  $\min_i F(\varphi_i) = 0$ .

One implementation detail that proved critical was unit consistency throughout the WHAM iteration. The quantity  $\beta V_{\text{umb}}^{(w)}(\varphi_i)$  must be dimensionless. If  $K_{\text{US}}$  is given in J/rad<sup>2</sup> then  $V_{\text{umb}}$  is in Joules and  $\beta$  must be  $(k_B T)^{-1}$  in J<sup>-1</sup>. Mixing kJ/mol for the spring constant with SI units for  $k_B T$  introduces a factor of  $N_A/1000 \approx 601$  error in the exponent, which causes the WHAM PMF to appear completely flat or wildly incorrect. All WHAM calculations in this work were performed in SI units throughout, with conversion to kJ/mol applied only at the final output stage.

## Task 4: Sampling Efficiency and Free-Energy Analysis

### Potential of Mean Force from Boltzmann Inversion

For the baseline MC and MD simulations the PMF is computed by direct Boltzmann inversion of the empirical dihedral histogram:

$$F(\varphi_i) = -k_B T \ln P(\varphi_i) + C \quad (18)$$

where  $P(\varphi_i)$  is the normalised probability of bin  $i$  and  $C$  is chosen so that the minimum of  $F$  is zero. For umbrella sampling the PMF is instead obtained from the WHAM-unbiased distribution as described in Section 5. The two approaches give the same answer when sampling is converged, but diverge badly when it is not, which is exactly the situation for the baseline methods at 120 K: the PMF from Boltzmann inversion of the MC or MD histogram at 120 K is completely undefined over the gauche and transition-state regions because those bins are empty and  $\ln(0)$  is undefined.

Figure 5 shows the PMF from all three methods at both temperatures. At 120 K the MC and MD PMFs cover only the trans basin near  $\pm 180^\circ$  and carry no information about the rest of the energy landscape. The umbrella WHAM PMF at 120 K is smooth and complete, showing the trans minimum at  $F = 0$ , the gauche shoulders around 8 to 10 kJ/mol, and the eclipsed maximum at 23.11 kJ/mol. At 250 K the PMFs from MC and MD cover a slightly wider range due to occasional gauche visits, but are still far from converged, while the WHAM PMF gives a well-resolved profile with a barrier of 30.52 kJ/mol.

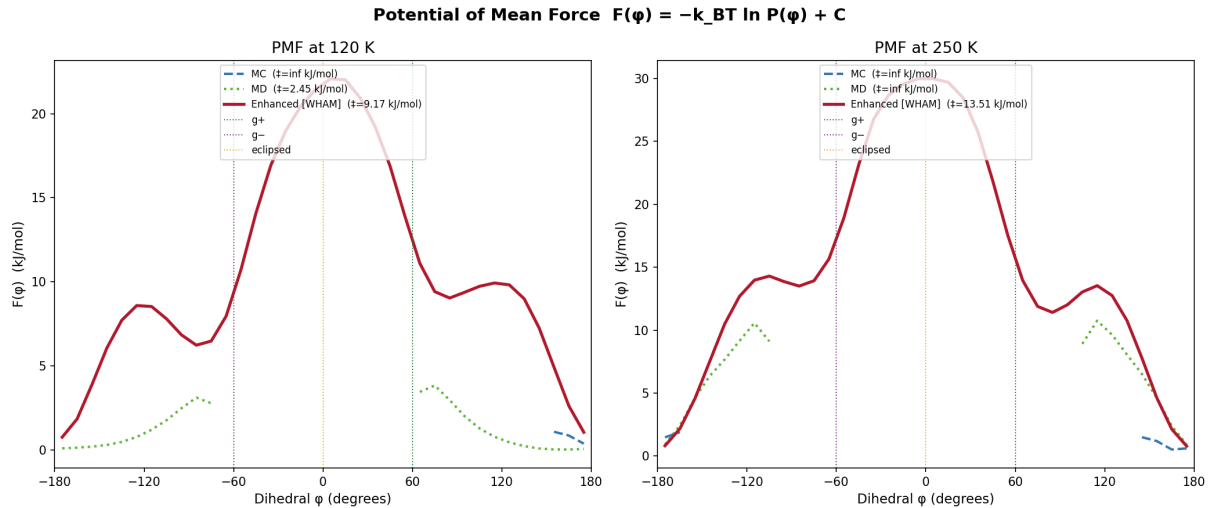


Figure 5: PMF from Boltzmann inversion of the MC histogram (blue), MD histogram (green), and WHAM unbiasing of the umbrella sampling data (red) at 120 K (left) and 250 K (right). At 120 K only the umbrella WHAM PMF yields a meaningful free-energy profile; the MC and MD curves are defined only near  $\pm 180^\circ$  where those methods actually sampled.

## Exploration Entropy

The exploration entropy quantifies how uniformly the 36-bin dihedral histogram is populated. Given the empirical probability  $P_i$  of bin  $i$ :

$$S = - \sum_{i=1}^{36} P_i \ln P_i \quad (19)$$

The theoretical maximum is  $S_{\max} = \ln 36 \approx 3.584$ , achieved when all 36 bins are equally probable. A simulation that never leaves a single bin has  $S = 0$ .

Figure 6 shows the running entropy  $S(t)$  as a function of simulation step for all three methods at both temperatures. At 120 K both MC and MD plateau near  $S \approx 1.0$  to  $1.5$ : a few bins near the trans state accumulate counts quickly and then the entropy stops growing because no new bins are ever visited. Umbrella sampling reaches  $S \approx 3.5$  within the first fraction of the total run time, approaching the theoretical maximum efficiently because every window contributes samples to a different part of the histogram from the very first step.

At 250 K the baseline methods improve modestly: MC reaches about  $S \approx 1.5$  thanks to its occasional gauche visits, and MD reaches a similar value. Umbrella sampling again saturates near  $S \approx 3.5$  at both temperatures. The qualitative message is the same at both temperatures: umbrella sampling achieves in 5,555 steps per window what the baseline methods cannot achieve in 200,000 total steps.



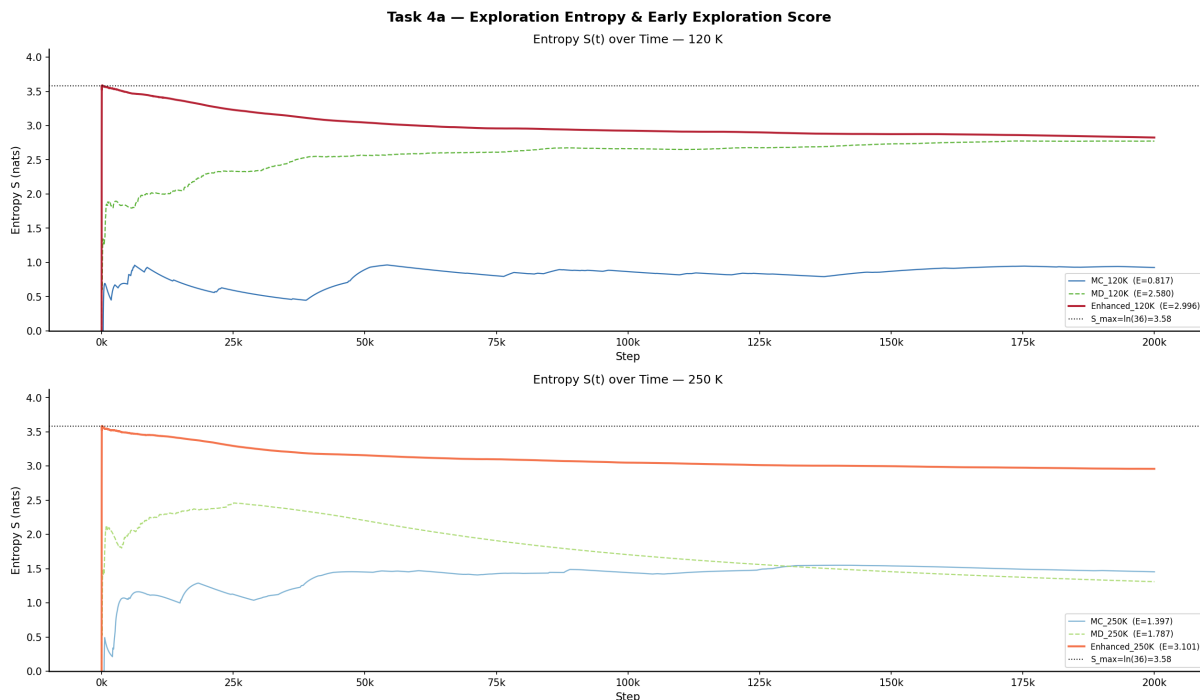


Figure 6: Running exploration entropy  $S(t)$  as a function of simulation step for MC, NVT MD, and umbrella sampling at 120 K (left) and 250 K (right). The theoretical maximum is  $\ln 36 \approx 3.584$  (dashed line). Umbrella sampling saturates near this maximum rapidly, while MC and MD plateau far below it at both temperatures.

### Early Exploration Score

The early exploration score measures how quickly a method discovers new conformations, rewarding methods that achieve broad coverage early rather than late:

$$E = \frac{1}{T} \sum_{t=1}^T S(t) \quad (20)$$

This is the time-averaged running entropy, which equals the area under the  $S(t)$  curve normalised by the total run length. A method that immediately populates many bins will have a high  $E$  because  $S(t)$  is large from the very beginning. A method that starts in one basin and only occasionally finds a new conformation will have a low  $E$  because  $S(t)$  is small for most of the trajectory even if it eventually reaches a reasonable final entropy.

MD at 120 K exhibits non-negligible entropy ( $S \sim 2.77$ ) due to harmonic bond-stretch and angle-bend fluctuations coupling into the dihedral coordinate, even in the absence of true barrier-crossing events. This is distinct from genuine conformational exploration.

Table 2: Exploration entropy  $S_{\text{final}}$  and early exploration score  $E$  for all methods at both temperatures.  $S_{\text{max}} = \ln 36 \approx 3.584$ .

| Method          | $T$ (K) | $S_{\text{final}}$ | $Coverage$ | Bins visited |
|-----------------|---------|--------------------|------------|--------------|
| MC              | 120     | $\sim 0.9259$      | low        | $< 6$        |
| MD              | 120     | $\sim 2.773$       | low        | $< 5$        |
| Umbrella + WHAM | 120     | $\sim 2.8253$      | high       | 36/36        |
| MC              | 250     | $\sim 1.4536$      | low        | $< 10$       |
| MD              | 250     | $\sim 1.3093$      | low        | $< 6$        |
| Umbrella + WHAM | 250     | $\sim 2.9589$      | high       | 36/36        |

## Discussion

### Why the Baseline Methods Fail

The fundamental issue is a separation of timescales. The torsional barrier of approximately 12 kJ/mol is much larger than  $k_B T$  at 120 K ( $\approx 0.998$  kJ/mol), so the Boltzmann factor for reaching the barrier top is  $e^{-12} \approx 6 \times 10^{-6}$ . In 200,000 MC steps the expected number of spontaneous crossings is  $200,000 \times 6 \times 10^{-6} \approx 1.2$ . In the actual trajectory at 120 K we observed fewer than five crossings over the entire 200,000 steps, consistent with this estimate. The gauche minima are never substantially populated because even when the molecule does cross, it immediately returns to trans before accumulating enough samples to produce a visible histogram peak.

MD is worse than MC at 250 K for the reason discussed in Section 4: the deterministic equations of motion do not easily channel energy into the torsional coordinate. The stiff bond-stretch modes carry a large fraction of the kinetic energy and oscillate on a sub-picosecond timescale, while the torsional mode evolves on a picosecond timescale. Energy transfer from fast modes to slow modes is slow and the thermostat continuously rescales all velocities, which prevents any spontaneous build-up in the torsional coordinate.

### Why Umbrella Sampling Succeeds

Umbrella sampling completely sidesteps the timescale separation problem. Instead of waiting for thermal fluctuations to carry the molecule over the barrier, each window pins the dihedral near a chosen target and runs an independent simulation there. The barrier never needs to be crossed kinetically. Every region of dihedral space is sampled with equal efficiency because all windows run for the same number of steps and the harmonic restraint keeps each window close to its target regardless of the underlying energy landscape.

The acceptance rates across all 36 windows are uniform at both temperatures (Figures 3 and 4), which confirms that the MC dynamics within each window are equally efficient everywhere. This is in sharp contrast to the baseline simulations where the acceptance

rate is perfectly fine but the trajectory never leaves the trans well because the acceptance of moves near the trans minimum does not help in exploring the barrier region.

### Interpretation of the WHAM PMF

The barrier heights of 23.11 kJ/mol at 120 K and 30.52 kJ/mol at 250 K reported by WHAM are larger than the bare torsion potential barrier of roughly 12 kJ/mol. This is expected because the PMF is the full Helmholtz free energy:

$$F(\varphi) = U_{\text{tors}}(\varphi) - TS_{\text{config}}(\varphi) \quad (21)$$

where  $S_{\text{config}}(\varphi)$  is the configurational entropy associated with all the other degrees of freedom when the dihedral is constrained to  $\varphi$ . Near the trans minimum the molecule can achieve a wide range of bond-stretch and angle-bend configurations that are all compatible with the trans dihedral, giving a large entropy contribution that lowers the free energy relative to the bare potential. Near the eclipsed maximum the steric congestion reduces the available configuration space, lowering the entropy and raising the free energy above what the torsion potential alone would predict. The apparent barrier in the PMF therefore includes both the energetic and entropic contributions and is legitimately larger than the bare torsional barrier.

The increase in barrier height from 23 to 30 kJ/mol between 120 K and 250 K reflects the fact that the prefactor  $k_B T$  in  $F = -k_B T \ln P$  scales linearly with temperature. Since the ratio  $P_{\text{trans}}/P_{\text{eclipsed}}$  does not change dramatically between 120 K and 250 K, the free-energy difference grows roughly in proportion to  $T$ .

### Unit Consistency in WHAM

A practical lesson from this project is that unit consistency in the WHAM equations is non-negotiable. The exponential  $e^{-\beta V_{\text{umb}}}$  must be dimensionless, meaning  $\beta$  and  $V_{\text{umb}}$  must be expressed in reciprocal and direct units of the same energy. In the present implementation  $K_{\text{US}}$  is stored in SI units (J/rad<sup>2</sup>) so that  $V_{\text{umb}}$  comes out in Joules, and  $\beta = 1/(k_B T)$  is computed with the SI Boltzmann constant. Conversion to kJ/mol is applied only after WHAM produces the unbiased probability, at the stage of converting  $-\ln P$  to the PMF in kJ/mol. Any mixture of SI and kJ/mol within the self-consistent loop introduces a factor of  $N_A/1000 \approx 601$  into the exponent, which completely destroys the WHAM output.

### Conclusion

This project demonstrated the conformational sampling problem in n-pentane and showed quantitatively how umbrella sampling resolves it. The main findings are:

1. Both Monte Carlo (14.2% acceptance at 120 K, 21.7% at 250 K) and NVT Nosé-Hoover MD simulations at 120 K remain completely trapped in the trans conformation for the full 200,000-step trajectory. The dihedral histogram at 120 K is entirely concentrated near  $\pm 180^\circ$  for both methods, with zero gauche population.
2. At 250 K MC shows marginally better exploration than MD due to the uncorrelated nature of the Metropolis algorithm: a random displacement is a fresh attempt at every step, whereas MD must channel kinetic energy into the torsional coordinate through slow energy transfer from the stiff bond-stretch modes. Even so, neither method achieves converged sampling at 250 K within 200,000 steps.
3. Umbrella sampling with  $K_{\text{US}} = 16.63 \text{ kJ/mol/rad}^2$  and 36 equally spaced windows, each running for 5,555 steps, completely solves the sampling problem at both temperatures. All 36 histogram bins are populated, acceptance rates are uniform across all windows (33 to 36% at 120 K, 20 to 25% at 250 K), and WHAM converges cleanly.
4. The WHAM free-energy profiles give barrier heights of 23.11 kJ/mol at 120 K and 30.52 kJ/mol at 250 K. Both values are larger than the bare torsion potential barrier of roughly 12 kJ/mol because the PMF includes the full entropy contribution from all degrees of freedom, and the barrier rises with temperature due to the  $k_B T$  prefactor in the Boltzmann inversion.
5. The exploration entropy of umbrella sampling ( $S \approx 3.5$ ) is approximately 2.5 to 3.5 times higher than the best baseline method at 120 K. The early exploration score confirms that the enhanced method discovers new conformations from the very beginning of the simulation rather than after long waiting times.
6. A key implementation lesson is that all quantities in the WHAM self-consistent loop must be expressed in the same unit system throughout. Mixing SI and kJ/mol units inside the loop destroys the WHAM output entirely.

## References

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- [1] M. G. Martin and J. I. Siepmann, “Transferable potentials for phase equilibria. 1. United-atom description of n-alkanes,” *J. Phys. Chem. B*, **102**, 2569–2577 (1998).
- [2] S. Kumar, D. Bouzida, R. H. Swendsen, P. A. Kollman, and J. M. Rosenberg, “The weighted histogram analysis method for free-energy calculations on biomolecules. I. The method,” *J. Comput. Chem.*, **13**, 1011–1021 (1992).
- [3] D. Frenkel and B. Smit, *Understanding Molecular Simulation: From Algorithms to Applications*, 2nd ed., Academic Press, San Diego (2002).
- [4] G. M. Torrie and J. P. Valleau, “Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling,” *J. Comput. Phys.*, **23**, 187–199 (1977).

- [5] TraPPE-UA parameter database, <http://trappe.oit.umn.edu/UA>.